

Exercises Set: Binary Classification

DSBA Mathematics Refresher 2026

Abstract

As this is the last session, there will be no compulsory questions this time.

1 Support Vector Machines



1.1 Theory

Define a line in $\mathbb{R}^2$ with parameters $\vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ and b ¹ defined by $\vec{w} \cdot \vec{x} = b$ (or $\vec{w} \cdot \vec{x} - b = 0$) for $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$. This line cut the plane in 2 regions:

- $\mathcal{R}_{-1}$: $\vec{x} \in \mathbb{R}^2$ such that $\vec{w} \cdot \vec{x} - b < 0$
- $\mathcal{R}_{+1}$: $\vec{x} \in \mathbb{R}^2$ such that $\vec{w} \cdot \vec{x} - b > 0$

The goal is to find $\vec{w}$ and b such that all points of the first class are in the first region, and all points of the second class are in the second region.

Let our data be $\{(\vec{x}_k, y_k)\}_{k=1}^N$ where $\vec{x}_k$ is the coordinate of the point in $\mathbb{R}^2$ and y_k is the label (+1 or -1).

The best line is not only separating the data in two sets, but also maximizing the distance between the line and the points.

1. Calculate the distance between the line and a point $\vec{x} \in \mathbb{R}^2$.

Let $h(\vec{x}) = \vec{w}^T \cdot \vec{x} - b$. We will predict first class -1 if $h(\vec{x}) < 0$ (as this means $\vec{x} \in \mathcal{R}_{-1}$); and second class +1 if $h(\vec{x}) > 0$ (as this means $\vec{x} \in \mathcal{R}_{+1}$). To have each item of data classified correctly, we need:

- if $y_k = +1$ then $h(\vec{x}_k) > 0$
- if $y_k = -1$ then $h(\vec{x}_k) < 0$

2. Derive a condition that ensures correct classification if true for all k .

Assuming the data is linearly separable, we now define 3 regions:

¹Here, $\vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, and b is a scalar. We use this notation because it makes it very easy to extend the technique to higher dimensions. For example, you can just add one component to use a plane to separate points in $\mathbb{R}^3$

- $\mathcal{R}_{-1}$: $\vec{x} \in \mathbb{R}^2$ such that $\vec{w} \cdot \vec{x} - b < -1$
 - $\mathcal{R}_0$: $\vec{x} \in \mathbb{R}^2$ such that $|\vec{w} \cdot \vec{x} - b| < 1$ (the "margin band")
 - $\mathcal{R}_{+1}$: $\vec{x} \in \mathbb{R}^2$ such that $\vec{w} \cdot \vec{x} - b > 1$
3. Calculate the margin band width.
 4. Formulate the constrained optimization problem to find the best (i.e. maximizing the margin) parameters $\vec{w}$ and b .

1.2 Practice

The training dataset consists of the following data points:

Positive class ("1"):

- (2, 2)
- (1, 1)

Negative class ("-1"):

- (0, 1)
- (1, 0)

1. Plot the points in a 2D graph.
2. Using the graph, find the optimal $\vec{w}$.
3. Calculate the optimal value of b to separate the data points while maximizing the margin.
4. Determine the equation of the optimal hyperplane in the form $\vec{w} \cdot \vec{x} + b = 0$.
5. Identify the support vectors in the dataset.
6. Calculate the margin, which is the perpendicular distance from the hyperplane to the nearest support vector.
7. Classify a new data point, (3, 2), based on the learned SVM model.

Remarks:

Here, we have made educated guesses to find $\vec{w}$ and b ; in practice, we need to solve the constrained optimization problem stated above.

This can be done using Lagrange multiplier, the details are outside the scope of this course.

In practice, we let computers solve the problem for us.

1.3 Kernel

Suppose we have the following data: Positive class ("1"):

- (0, 1)
- (0, -1)
- (1, 0)
- (-1, 0)

Negative class ("-1"):

- (0, 2)
- (0, -2)
- (2, 0)

- $(-2, 0)$
1. Plot the data points.
Observe that there is no line separating the two classes perfectly.

Let $\phi(x, y) = (x^2 + y^2, x^2 - y^2)$.

2. Compute $\phi(x, y)$ for every point, and plot them on a new graph.
Observe that the two classes should now be separable by a line ².

2 Logistic Regression

The SVM answers with a hard "yes" or "no". Often we would rather have a **probability**: not "this student will pass", but "this student passes with probability 0.8". This is what logistic regression gives us.

Throughout this section, we work with the following data: the number of hours a student has studied, and whether they passed the exam.

Hours Studied (X)	Pass (Y)
0	0 (no)
4	0 (no)
3	1 (yes)
8	1 (yes)

The model we want to fit is

$$P(Y = 1 \mid X = x) = \sigma(\beta_0 + \beta_1 x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

where σ is the **sigmoid** function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

2.1 The sigmoid function

Before fitting anything, let us get familiar with σ .

1. Compute $\sigma(0)$, $\sigma(2)$ and $\sigma(-2)$.
2. What are the limits of $\sigma(z)$ as $z \rightarrow +\infty$ and as $z \rightarrow -\infty$? Deduce that σ takes its values in $(0, 1)$ — which is what allows us to read it as a probability.
3. Show that $\sigma(-z) = 1 - \sigma(z)$.
Hint: multiply the numerator and the denominator of $\sigma(-z)$ by e^{-z} .
4. Show that

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

This surprisingly simple derivative is the reason logistic regression is so pleasant to optimize; we will use it in Section 2.4.

5. Sketch the graph of σ . For which value of z does the model hesitate between the two classes?

²The "kernel trick" allows us to perform an SVM on data transformed by a non-linear function; the details are outside the scope of this course.

2.2 Odds and log-odds

Gamblers do not speak of the probability p of an event, but of its **odds** $\frac{p}{1-p}$ (a probability of 0.75 means odds of 3, i.e. "3 to 1").

1. Let $p = \sigma(z)$. Show that

$$\frac{p}{1-p} = e^z \quad \text{and therefore} \quad \log\left(\frac{p}{1-p}\right) = z$$

2. Deduce that, for our model,

$$\log\left(\frac{P(Y=1 | X=x)}{P(Y=0 | X=x)}\right) = \beta_0 + \beta_1 x$$

In words: logistic regression is a *linear* model — not of the probability, but of the **log-odds**.

3. Suppose a student studies one hour more. By what factor are their odds of passing multiplied? (Your answer should depend on β_1 only.)

This is how logistic regression is read in practice: the coefficient β_1 is not a change in probability, it is a multiplier on the odds.

2.3 Measuring how good a model is

Take, for now, the (arbitrary) parameters

$$\beta_0 = -2 \quad \text{and} \quad \beta_1 = 0.6$$

1. Compute the four predicted probabilities $p_k = P(Y=1 | X=x_k)$ for the four students of the table.
2. If we predict "pass" whenever $p_k > 0.5$, how many of the four students are correctly classified?
3. The **log loss** of a single data point is

$$\ell(x_k, y_k) = -(y_k \log(p_k) + (1 - y_k) \log(1 - p_k))$$

Explain why, since y_k is either 0 or 1, this is always equal to

$$-\log(\text{the probability the model gave to the } \textit{correct} \text{ answer})$$

4. Compute the log loss of each of the four points. Which point costs the most? Does that match your intuition?
5. The **total cost** is $\mathcal{L}(\beta_0, \beta_1) = \sum_{k=1}^4 \ell(x_k, y_k)$. Compute it.
6. What would the total cost be for a model that predicts $p = \frac{1}{2}$ for everybody (i.e. $\beta_0 = \beta_1 = 0$)? Is our model better than that?
7. What happens to the log loss of a point if the model gives probability 0 to the correct answer? Why is this a desirable property for a cost function?

2.4 Fitting the model

We now look for the β_0 and β_1 that make the total cost as small as possible. Unlike the linear regression of Session 3, there is no formula for the answer: we must use gradient descent.

1. Using $\sigma' = \sigma(1 - \sigma)$, show that for a single data point

$$\frac{\partial \ell}{\partial \beta_0} = p_k - y_k \quad \text{and} \quad \frac{\partial \ell}{\partial \beta_1} = x_k (p_k - y_k)$$

and deduce the two partial derivatives of the total cost $\mathcal{L}$.

Remark: the quantity $p_k - y_k$ is simply the error made on the k -th point. It is hard to imagine a simpler gradient.

2. Write the two update rules of gradient descent for β_0 and β_1 , with a learning rate η .
3. Starting from $\beta_0 = \beta_1 = 0$ and with $\eta = 0.05$, perform **one** step by hand. Comment on the sign of the change of β_1 .
4. Using a computer, iterate until convergence and give the fitted β_0 and β_1 , and the minimal cost. (A notebook is provided; you may also use the grid search of Session 3 to get a first rough answer.)
5. At the minimum, the gradient vanishes. Show that this means

$$\sum_k p_k = \sum_k y_k \quad \text{and} \quad \sum_k x_k p_k = \sum_k x_k y_k$$

and use these two identities to check your numerical answer.

2.5 Reading the fitted model

1. Estimate the probability of passing for a student who studied 0, 1 and 10 hours.
2. Above how many hours of study does the model predict a pass? (This is the **decision boundary**; express it in terms of β_0 and β_1 before computing it.)
3. Using Section 2.2, give the factor by which one extra hour of study multiplies the odds of passing.
4. Can this model classify all four students of the table correctly? Look again at the data before answering, and explain what this implies for the minimal cost you found.
5. Would you trust these numbers to advise a real student?